

$$\boxed{3.9} \quad (1) \quad 1 - \frac{3}{8} \times \frac{2}{7} = \frac{25}{28}$$

$$(2) \quad \frac{\frac{3}{8} \times \frac{2}{7}}{\frac{3}{8} \times \frac{2}{7} + \frac{5}{8} \times \frac{3}{7}} = \frac{6}{6+15} = \frac{2}{7}$$

$$\boxed{3.10} \quad (1) \quad \frac{5}{5+4} = \frac{5}{9}$$

$$(2) \quad \frac{5 \times 4}{5 \times 11} = \frac{4}{11}$$

$$(3) \quad \frac{5 \times 4 \times 3}{5 \times 11 \times 10} = \frac{6}{55}$$

$$\boxed{3.11} \quad P(A) = \frac{1}{1000}, P(B|A) = \frac{99}{100}, P(B|A^c) = \frac{1}{10}$$

$$(1) \quad P(B) = P(A)P(B|A) + P(A^c)P(B|A^c)$$

$$= \frac{1}{1000} \times \frac{99}{100} + \frac{999}{1000} \times \frac{1}{10}$$

$$= \frac{10089}{100000}$$

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} = \frac{99}{10089} \approx 0.98\%$$

$$(2) \quad P(C) = P(A)P(C|A) + P(A^c)P(C|A^c)$$

$$= \frac{1}{1000} \times \frac{99}{100} \times \frac{99}{100} + \frac{999}{1000} \times \frac{1}{10} \times \frac{1}{10}$$

$$= \frac{109701}{10000000}$$

$$P(A|C) = \frac{9801}{109701} \approx 8.93\%$$

$$\begin{array}{r} 99 \\ \times 99 \\ \hline 891 \\ 8910 \\ \hline 9801 \\ \times 999 \\ \hline 99900 \\ 999000 \\ \hline 109701 \end{array}$$

$$\boxed{3.12} \quad (1) \quad P(A) = \frac{1}{200}, P(B|A) = \frac{99}{100}, P(B^c|A^c) = \frac{99}{100} \Leftrightarrow P(B|A^c) = \frac{1}{100}$$

$$P(A \cap B) = P(A)P(B|A) = \frac{1}{200} \times \frac{99}{100} = \frac{99}{20000}$$

$$(2) \quad P(A^c \cap B) = P(A^c)P(B|A^c) = \frac{99}{200} \times \frac{1}{100} = \frac{99}{20000}$$

$$(3) \quad P(B) = P(A \cap B) + P(A^c \cap B) = \frac{99}{10000}$$

$$P(A^c|B) = \frac{P(A^c \cap B)}{P(B)} = \frac{99/20000}{99/10000} = \frac{1}{2}$$

$$(4) \quad P(B^c) = \frac{9901}{10000}$$

$$P(A|B^c) = \frac{P(A)P(B^c|A)}{P(B^c)} = \frac{\frac{1}{200} \times \frac{1}{100}}{\frac{9901}{10000}} = \frac{1}{19802}$$

$$\begin{array}{r} 10000 \\ - 99 \\ \hline 9901 \\ + 2 \\ \hline 19802 \end{array}$$

$$\boxed{3.13} \quad P(A) = \frac{1}{10000}, P(B|A) = \frac{99}{100}, P(B|A^c) = \frac{1}{100}$$

$$(1) \quad P(B) = P(A)P(B|A) + P(A^c)P(B|A^c)$$

$$= \frac{1}{10000} \times \frac{99}{100} + \frac{9999}{10000} \times \frac{1}{100}$$

$$= \frac{10098}{1000000}$$

$$P(A|B) = \frac{99}{10098} \approx 0.98\%$$

(2) この検査にて陽性になったとしても、実際に病気として発症しない可能性が高いため、少なくとも陽性になった場合は信用できない。